Output:-

Que :- A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

Code:-

```
GNU nano 7.2
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12, frames = 3;
    int frame[3];
    int i, j, k, pos, hits = 0, faults = 0;

    for(i = 0; i < frames; i++)
        frame[i] = -1;

    printf("Page Reference String: ");
    for(i = 0; i < n; i++)
        printf("%d ", pages[i]);

    printf("\nNumber of Frames: %d\n\n", frames);

    for(i = 0; i < n; i++) {
        int found = 0;

        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        if(!found) {
            int farthest = i, index = -1;

            for(j = 0; j < frames; j++) {
                for(k = i+1; k < n; k++) {
                    if(frame[j] == pages[k])
                        break;
                }

                if(k == n) {
                    index = j;
                    break;
                }
```

```
GNU nano 7.2
                found = 1;
                break;
            }
        }

        if(!found) {
            int farthest = i, index = -1;

            for(j = 0; j < frames; j++) {
                for(k = i+1; k < n; k++) {
                    if(frame[j] == pages[k])
                        break;
                }

                if(k == n) {
                    index = j;
                    break;
                }

                if(k > farthest) {
                    farthest = k;
                    index = j;
                }
            }

            if(index == -1)
                index = 0;

            frame[index] = pages[i];
            faults++;
        }
    }

    printf("\nPage Hits = %d\n", hits);
    printf("Page Faults = %d\n", faults);
    printf("Hit Ratio = %.2f%%\n", (hits/(float)n)*100);
    printf("Fault Ratio = %.2f%%\n", (faults/(float)n)*100);

    return 0;
```

Output:-

```
MINGW64:/c/Users/ASUS/OneDrive/Desktop/OSLab_CM24100/practical 8
./optimal
gcc.exe: error: optimal.c: No such file or directory
gcc.exe: fatal error: no input files
compilation terminated.
bash: ./optimal: No such file or directory

ASUS@LAPTOP-4RQNFJOR MINGW64 ~/OneDrive/Desktop/OSLab_CM24100 (main)
$ cd "practical 8"

ASUS@LAPTOP-4RQNFJOR MINGW64 ~/OneDrive/Desktop/OSLab_CM24100/practical 8 (main)
$ gcc optimal.c -o optimal

ASUS@LAPTOP-4RQNFJOR MINGW64 ~/OneDrive/Desktop/OSLab_CM24100/practical 8 (main)
$ ./optimal
Page Reference String: 1 2 3 4 1 2 5 1 2 3 4 5
Number of Frames: 3

Page Hits = 5
Page Faults = 7
Hit Ratio = 41.67%
Fault Ratio = 58.33%

ASUS@LAPTOP-4RQNFJOR MINGW64 ~/OneDrive/Desktop/OSLab_CM24100/practical 8 (main)
$
```